

MORITZWECKBECKER

AI safety researcher focused on interpretability and alignment of frontier models. Oxford and Cambridge mathematics graduate, pursuing a PhD at Fraunhofer HHI. Research Interests include data attribution, subliminal learning, and mechanistic interpretability.

Personal Details

Date / Place of Birth	17/01/1999 / Berlin
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Education

01/2024 – Fraunhofer HHI & TU Berlin	PhD in AI Interpretability (under Prof. Wojciech Samek) Data Attribution for Deep Learning models, Subliminal Learning
10/2021 – 09/2022 University of Oxford	Master of Science, Statistical Science / Pass with distinction (top-ranked graduate) Statistical Data Analysis, Statistical Modelling, Machine Learning, Deep Learning
10/2020 – 06/2021 University of Cambridge	Master of Advanced Study, Pure Mathematics / Pass with distinction (top 15%) Probability Theory, Medical Statistics, Causal Inference
10/2016 – 08/2019 TU Berlin	Bachelor of Science, Mathematics / 1.0 (highest possible grade) Algebra, Analysis, Programming, Numerical Mathematics

Publications

Safety & Alignment

- Weckbecker, M.**, Müller, J., Hagag, B., & Mulet, M. (2026). **Thought Virus: Viral Misalignment via Subliminal Prompting in Multi-Agent Systems.** *(under review at COLM and AIWILD workshop at ICML)*
- Jaburi, L., Paulo, G., Quirke, L., Shabalin, S., Mulet, M., Müller, J., Jena, S., **Weckbecker, M.**, Belrose, N. *(in preparation)*. **Mitigating Emergent Misalignment with Data Attribution.**

Data Attribution

- Yolcu, G. Ü.*, **Weckbecker, M.***, Wiegand, T., Samek, W., & Lapuschkin, S. (2025). **DualXDA: Towards Sparse, Efficient and Explainable Data Attribution in Large AI Models.** *Transactions on Machine Learning Research.*
- Naujoks, J. R., Krasowski, A., **Weckbecker, M.**, ..., Klausen, R. P. (2025). **Leveraging Influence Functions for Resampling Data in Physics-Informed Neural Networks.** *World Conference on Explainable Artificial Intelligence.*
- Naujoks, J. R., Krasowski, A., **Weckbecker, M.**, ..., & Klausen, R. P. (2024). **PINNfluence: Influence Functions for Physics-Informed Neural Networks.** *2024 NeurIPS Workshop: Machine Learning and the Physical Sciences.*

Other ML Projects

- Weber, L., Berend, J., **Weckbecker, M.**, ..., Lapuschkin, S. (2025). **Efficient and flexible neural network training through layer-wise feedback propagation.** *Transactions on Machine Learning Research.*
- Pahde, F., Dreyer, M., Weber, L., **Weckbecker, M.**, Anders, C. J., Wiegand, T., ... & Lapuschkin, S. (2025). **Navigating neural space: Revisiting concept activation vectors to overcome directional divergence.** *ICLR.*

- **Weckbecker, M.,** Anžel, A., Yang, Z., & Hattab, G. (2024). **Interpretable molecular encodings and representations for machine learning tasks.** *Computational and Structural Biotechnology Journal*.
- **Weckbecker, M.,** Neziri, S., & Hattab, G. (2023). **Supporting Mathematical Education with Interactive Visual Proofs.** *2023 IEEE Workshop on Visualization for Social Good (VIS4Good)*.
- **Weckbecker, M.,** Xu, W. & Reinert, G. (2022). **On RKHS Choices for Assessing Graph Generators via Kernel Stein Statistics.** *2022 NeurIPS Workshop on Score-Based Methods*.

Work Experience

01/2023 – 09/2023 Robert Koch-Institute	Researcher in AI for Public Health • Developed interpretable molecular encoding method (iCAN) for peptide and protein classification in biomedical applications • Outperformed state-of-the-art encodings on large-scale study across 50 datasets, published in Computational and Structural Biotechnology Journal
12/2021 University of Oxford	Research Assistant Intern at the School of Geography and the Environment • Data analysis and visualization of big data networks on offshore jurisdiction and tax havens
06/2020 – 09/2020 OmicEra Diagnostics	Python Developer • Programming new algorithm for protein sequencing using graph theory • Data analysis in Python using data from biological experiments • Improving the algorithm using Machine Learning approaches
04/2018 – 03/2019 TU Berlin	Academic Tutor • Teaching students in Linear Algebra and Analysis

Awards

09/2022	Gutiérrez Toscano Prize in Applied Statistics for best performance in the MSc in Statistical Sciences
12/2019	Award for the best presentation of a thesis by the Mathematics department at TU Berlin
11/2019	Bachelor Award from “Berliner Mathematische Gesellschaft”
11/2016	Scholarship from the German Academic Scholarship Foundation (top 0.25% of students) for bachelor's and master's degrees

Volunteering

04/2024 – 05/2025 be.boosted e.V. (Leadership Development Organisation)	Member of the Board • Lead team of 50+ active members and provided strategic oversight for leadership training programmes and alumni network (300+ members)
04/2023 – 04/2024 be.boosted e.V.	Head of Recruiting • Led team of 10 members and 20+ volunteers to attract 250+ applications and select annual fellowship cohort
04/2021 – 04/2023 be.boosted e.V.	Trainer / Senior Trainer • Prepared and facilitated weekly interactive workshops on communication, speechmaking and group dynamics • Led in-person weekend seminar and mentored participants through Model UN simulations and debate competitions • Trained and supported new trainers in leadership development methodologies
09/2018 – 09/2020 TU Berlin	Teacher for gifted students at the mathematical association “Leonhard Euler” • Taught advanced mathematics to high-performing young students; developed curriculum materials, lessons plan and problem sets